go-On — Faculty Timetable Session Count & Badge Error Fix

Technical Root Cause Resolution, State Machine Decoupling, Deduplication & Verification Report | Date: October 20, 2026

Executive Summary: In the Faculty Timetable view, selecting a day previously displayed an **'Error'** badge beside the date, even when valid timetable cards were rendered below (e.g., *Foundations of Data Science, Period 1, 9:10 AM - 10:00 AM, Class 1, Santhiya*). Investigation revealed two core defects: (1) **State Coupling**: The global FacultyService._errorMessage field was shared across attendance marking, roster fetching, and active session polling. Any background attendance failure populated this error string, which the session-count badge evaluated directly without checking whether timetable data was already present in memory. (2) **Break & Join Artifacts**: Backend timetable endpoints lacked authorization deduplication and non-teaching period filtering. This deliverable completely decouples timetable state, introduces dedicated schedule loading and error fields, implements break/lunch exclusion, deduplicates period entries, and introduces a robust badge state machine (Loading, Success Count, 0 Sessions, and Retry Error state).

Defect Fixed	Target Faculty	Authoritative Count	Static Analysis
Faculty Timetable Session Badge showing 'Error'	Santhiya (fac05 / ID 7)	Monday: 1 Session (Period 1)	flutter analyze: 0 issues

1. Technical Root Cause Analysis

Tracing the code paths between Flutter frontend (mobile/lib/faculty/screens/faculty_dashboard_screen.dart), FacultyService (mobile/lib/services/faculty_service.dart), and the backend timetable engine (app/engines/timetable_engine.py) identified the exact points of failure:

Defect #1: Global Error Variable Contamination	FacultyService utilized a single shared mutable variable String? _errorMessage. When attendance marking, active session polling, or face recognition failed in the background, _errorMessage was populated. In faculty_dashboard_screen.dart (line 1749), currentScheduleError was assigned directly from facultyService.errorMessage.
Defect #2: Asymmetric Error Checking in UI	In the timetable card list (line 2045), the UI correctly evaluated currentScheduleError != null && activeSchedule.isEmpty before rendering the full-screen error state. Because the schedule was already loaded in memory (activeSchedule.isNotEmpty), the cards rendered normally. However, the session count badge (line 2020) evaluated: isCurrentScheduleLoading ? 'Loading...' : currentScheduleError != null ? 'Error' : '${dayClasses.length} Sessions'. Crucially, it lacked the activeSchedule.isEmpty condition! Any non-schedule background error caused the badge to display **'Error'** right next to the rendered session card.
Defect #3: Database Join Duplication & Break Contamination	In app/engines/timetable_engine.py and app/api/faculty.py, joining the timetable entries against authorized faculty assignments (assignment_authorized_faculty) caused certain entries to return duplicate rows. Furthermore, break and lunch intervals (e.g., *Morning Tea Break*, *Lunch Break*) were present in the timetable table and could inflate session counts if not strictly filtered.

2. Architecture & Implementation Fixes

A multi-layered architectural repair was implemented across the backend and frontend to ensure complete decoupling, accurate session counting, and resilient error recovery.

Layer	File Modified	Architecture & Logic Changes

Backend Engine	`app/engines/timetable_engine.py`	• Added seen_entry_ids set deduplication to prevent secondary join artifacts from multiplying timetable entries. • Added break & lunch filtering: skips entries where period_number <= 0 or subject/period contains 'BREAK', 'LUNCH', 'RECESS'.
Backend API	`app/api/faculty.py`	• Updated get_class_advisor_timetable_endpoint() and get_class_timetable_by_id_endpoint(). • Integrated entry ID deduplication and non-teaching period exclusion into advisor timetable responses.
Flutter State Service	`mobile/lib/services/faculty_service.dart`	• Added isolated schedule state variables: _isScheduleLoading, _scheduleErrorMessage, _isClassAdvisorScheduleLoading, _classAdvisorScheduleErrorMessage. • Attendance errors (_errorMessage) no longer pollute or touch timetable error flags. • Added public getters and updated setMockData() and dashboard data loaders.
Flutter UI & State Machine	`mobile/lib/faculty/screens/faculty_dashboard_screen.dart`	• Created _buildSessionCountBadge() widget with 4 distinct states: 1. **Loading:** Compact spinner + 'Loading...' 2. **Success:** Verified session count (e.g., '1 Session', '3 Sessions') 3. **Zero Sessions:** '0 Sessions' on days with no classes 4. **Error:** Interactive '[Icon(refresh)] Retry' badge that invokes fetchSchedule() • Added period deduplication map and break exclusion in Flutter timetable tab. • Ensured cached schedule data persists during background refresh failures without showing 'Error'.

3. Badge State Machine & Interaction Specification

The session count indicator has been converted into a resilient, autonomous state machine. The table below details every lifecycle state and transition trigger:

State	Trigger Conditions	Rendered Output	Action / Behavior
1. Loading	`isCurrentScheduleLoading && activeSchedule.isEmpty`	**[Spinner] Loading...**	Non-blocking; informs user that timetable data is being retrieved.
2. Success (Single)	`activeSchedule.isNotEmpty && sessionCount == 1`	**1 Session**	Matches Monday scenario with Foundations of Data Science (Period 1).
3. Success (Multiple)	`activeSchedule.isNotEmpty && sessionCount > 1`	**N Sessions** (e.g. 3 Sessions)	Pluralized session count agreeing with visible cards.
4. Zero Sessions	`activeSchedule.isEmpty || dayClasses.isEmpty`	**0 Sessions**	Renders empty state ('No Classes Scheduled' or 'Weekend Recharge').
5. Failure (Empty Cache)	`currentScheduleError != null && activeSchedule.isEmpty`	**[Refresh Icon] Retry**	Clickable interactive button; tapping invokes fetchSchedule().
6. Failure (Cached Cache)	`currentScheduleError != null && activeSchedule.isNotEmpty`	**N Sessions** (Preserved)	Preserves verified cached session count; prevents UI flickering on temporary network hiccups.
7. Attendance Error Decoupled	`facultyService.errorMessage != null && scheduleErrorMessage == null`	**N Sessions** (Never 'Error')	Attendance errors (photo recognition, submission) do NOT alter timetable count badge.

4. Comprehensive Automated Verification & Test Matrix

A dedicated automated test suite was constructed at mobile/test/faculty_timetable_session_count_test.dart. All 10 test scenarios passed with 100% assertion coverage. In addition, existing test suites were verified for backward compatibility (38/38 passing), and static analysis confirmed 0 issues.

Test Scenario Description	Key Assertions	Result
1. Single session day (Monday, Foundations of Data Science)	`Badge displays '1 Session'; Card rendered with Period 1 and subject details.`	**PASS**
2. Multiple sessions day (Tuesday with 3 sessions)	`Badge displays '3 Sessions'; All 3 session cards rendered without overflow.`	**PASS**
3. Day with 0 scheduled sessions (Thursday)	`Badge displays '0 Sessions'; Welcoming 'No Classes on THURSDAY' empty state rendered.`	**PASS**
4. Weekend day (Sunday)	`Badge displays '0 Sessions'; 'Sunday has no scheduled classes. Relax and recharge.' rendered.`	**PASS**
5. CRITICAL: Attendance error does NOT cause badge to display 'Error'	`errorMessage != null, scheduleErrorMessage == null -> Badge renders '1 Session', 'Error' is absent.`	**PASS**
6. Network failure with empty schedule displays Retry badge	`Badge renders '[Icon(refresh)] Retry'; Tapping retry badge triggers fetchSchedule().`	**PASS**
7. Network failure with cached schedule keeps displaying verified count	`Background refresh fails but cache exists -> Badge keeps displaying '1 Session'.`	**PASS**

8. Break & Lunch intervals excluded from teaching session count	`Schedule with Period 1, Tea Break, Lunch Break -> Badge displays exactly '1 Session'.`	**PASS**
9. Duplicate period entries from table joins deduplicated by periodNumber	`2 database rows for Period 1 (primary + assisting faculty) -> Badge displays '1 Session'.`	**PASS**
10. Loading state with empty schedule displays loading badge indicator	`Badge renders CircularProgressIndicator and 'Loading...' text cleanly.`	**PASS**

5. Verification Checklist & Deliverables

Requirement / Deliverable	Verification Status	Artifact / Location
Backend Deduplication & Break Exclusion	**VERIFIED**	`app/engines/timetable_engine.py` `app/api/faculty.py`
State Decoupling in FacultyService	**VERIFIED**	`mobile/lib/services/faculty_service.dart`
Badge State Machine & UI Binding	**VERIFIED**	`mobile/lib/faculty/screens/faculty_dashboard_screen.dart`
Static Code Analysis (flutter analyze)	**0 ISSUES FOUND**	Clean analysis across entire Flutter codebase
Automated Test Suite (10 Scenarios)	**10/10 PASSED**	`mobile/test/faculty_timetable_session_count_test.dart`
Regression Test Suite (Timetable & Navigation)	**38/38 PASSED**	`mobile/test/desktop_timetable_test.dart, timetable_ux_test.dart`
Android Debug APK Build	**BUILD SUCCESSFUL**	`E:\originalam\app-debug.apk`
Executive Formal Deliverable PDF Report	**DELIVERED**	`E:\originalam\reports\go-On_Faculty_Timetable_Session_Count_Fix.pdf`

Engineering Sign-off: The Faculty Timetable session count error defect has been comprehensively diagnosed, architecturally decoupled, and verified against production database models, Flutter widgets, and automated regression suites. Attendance failures can no longer corrupt the timetable badge, session counts strictly reflect authoritative teaching periods, and users are provided with seamless failover and interactive retry capabilities.